
DATA CO SUPPLY CHAIN ANALYTICS

End-to-End Analytics & Data Forensics Investigation

Report Title	DataCo Supply Chain Analytics
Subtitle	End-to-End Analytics & Data Forensics Investigation
Dataset	DataCo SMART Supply Chain (research dataset)
Analysis Window	2015-01-01 to 2017-09-30 (validated real-data period)
Tooling	PostgreSQL, SQL, Python (pandas, numpy), star schema warehouse
Audience	Recruiter / hiring manager (data analyst portfolio)
Report Type	Standalone end-to-end project report
Status	Complete

A complete analytical investigation documenting the full analyst workflow, warehouse construction, eight structured investigations, and a consolidated data forensics assessment of dataset integrity.

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SECTION 1 — EXECUTIVE SUMMARY

1.1 Purpose

This report documents a full end-to-end analytics investigation of the DataCo Supply Chain dataset. The project began as a business analysis exercise and evolved into a data forensics investigation when evidence emerged that the dataset is partially synthetic.

The report presents the complete work: business understanding, data discovery, warehouse construction, warehouse validation, eight structured business investigations, and a consolidated forensics assessment of the dataset's integrity.

1.2 What Was Done

The project followed a complete analytics workflow:

- Phase 1 Business Understanding
- Phase 2 Source System Discovery
- Phase 3 Data Dictionary
- Phase 4 Data Profiling
- Phase 5 Data Cleaning
- Phase 6 Business System & Data Modeling (star schema)
- Phase 7 PostgreSQL Analytical Warehouse
- Phase 8 Warehouse Validation (20 verification tests)
- Phase 9 Business Question Design (24 questions mapped)
- Phase 10 SQL Analytical Investigation (8 investigations)
- Phase 11 Data Forensics Investigation (9 findings)
- Phase 12 Final Report (this document)

The warehouse was built on a validated star schema: **analytics.dim_customer**, **analytics.dim_product**, **analytics.dim_date**, **analytics.fact_sales** (order-line-item grain, 180,519 rows). The warehouse passed all 20 validation checks: completeness, grain, dimension uniqueness, referential integrity, measure reconciliation, and date-range consistency.

1.3 What Was Found

The eight investigations produced descriptive findings from the dataset, and each was subsequently subjected to a forensics audit. The central finding of the project is:

The DataCo dataset is a research artifact, not recorded transactions from a real company. Portions of it are synthetic, and several dimensions are randomly assigned.

Nine forensics findings were confirmed:

F1. The dataset is a research artifact

Source: Kaggle, published for Big Data / ML research, not from a real company's operational systems.

F2. October 2017 onward is synthetic

Daily order counts fixed at 68/69 alternating; lines-per-order fixed at exactly 1.0; sales values repeat across days.

F3. Market labels are unstable across years

Entire market-years are missing (LATAM has no 2016 data; USCA has no 2015 data; etc.).

F4. Margins are uniformly ~10–11% across every dimension

Markets 10.12%–11.14%; Regions 9.64%–13.51%; Categories 9.90%–17.46%; Segments 10.70%–10.78%. No real business exhibits this uniformity.

F5. Categories are randomly assigned

Product names do not match their category labels (e.g., golf balls classified as "Electronics").

F6. order_profit_per_order appears to be order-level

The field is identical to benefit_per_order on every row and appears to repeat the same value across all line items of an order.

F7. A \$1,500 price anchor appears

P99 order value is exactly \$1,500.00, and the most extreme negative-margin lines cluster around \$1,500 product prices.

F8. Discount distribution is artificial

All 18 distinct discount rates appear approximately 9,550 times each. No real discount system behaves this way.

F9. Shipping data is synthetic

Standard Class shows zero day-variance across all 34,331 orders (real = scheduled = 4.00 days exactly). First Class has higher recorded late risk than Standard Class.

1.4 What This Means

The dataset cannot support real business insights about a company called DataCo. The "DataCo business" is a research construct.

The dataset **CAN** support: practicing the full analyst workflow; data forensics methodology demonstration; machine learning experimentation on structured data; education and benchmark testing.

The dataset **CANNOT** support: real business insights or strategic recommendations; market, category, or segment performance analysis; profitability analysis (given F6 uncertainty); growth-by-market analysis (given F3); discount strategy analysis (given F8); shipping performance analysis (given F9).

1.5 Report Structure

Sections 2–3 provide project context and methodology. Sections 4–11 present the eight investigations in full. Section 12 consolidates all forensics findings. Sections 13–14 define what the dataset can and cannot support. Section 15 concludes. Appendices provide the business question map, warehouse schema, and SQL script references.

End of Section 1. Proceed to Section 2: Project Overview.

SECTION 2 — PROJECT OVERVIEW

2.1 The Dataset

The DataCo Supply Chain dataset is publicly available on Kaggle under the title "DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS." It was published by Shashwat Tiwari of Instituto Politecnico de Leiria as a resource for Big Data and Machine Learning research. It is not a real company's operational data. It is a research artifact designed for experimentation and algorithm benchmarking.

Property	Value
Source file	DataCoSupplyChainDataset.csv
Rows	180,519
Columns (raw)	53
Columns (cleaned)	51
Primary key	Order Id + Order Item Id (composite)
Grain	Order-line item
Raw date range	2015-01-01 to 2018-01-31
Valid analysis window	2015-01-01 to 2017-09-30

2.2 The Goal

The original project goal was to build a supply chain analytics warehouse and answer real business questions about DataCo's performance across sales, profitability, customers, products, shipping, and regions. The goal evolved during the investigation. After repeated patterns of synthetic construction were detected, the project added a data forensics layer. The final goal became: reconstruct, model, validate, and investigate the dataset, then evaluate whether the analytical patterns it contains can legitimately support business interpretation.

2.3 The Scope

In scope: Full analytics workflow (Phases 1–12); star schema modeling; PostgreSQL warehouse construction; 20-test warehouse validation suite; eight structured SQL investigations; consolidated data forensics assessment.

Out of scope: Power BI dashboard (deferred); machine learning modeling (dataset not suitable for real-world pattern learning); real business recommendations (dataset does not support them).

2.4 What This Report Is

A complete record of the analytical investigation; a documentation of every phase of the workflow; a forensics audit of dataset integrity; an honest conclusion about what the dataset can and cannot support.

2.5 What This Report Is Not

Not a set of business insights about a real company; not a dashboard or executive summary for DataCo executives; not a claim that the dataset describes reality. The report's honesty about this distinction is one of its strengths. Most analytical projects present findings without questioning whether the data deserves interpretation. This project does both.

End of Section 2. Proceed to Section 3: Methodology.

SECTION 3 — METHODOLOGY

3.1 Analytical Approach

The project followed a structured, phased methodology. Each phase produced a durable artifact that fed the next phase. No phase was skipped, and no shortcut was taken. The twelve phases are: Business Understanding; Source System Discovery; Data Dictionary; Data Profiling; Data Cleaning; Business System & Data Modeling; PostgreSQL Warehouse; Warehouse Validation; Business Question Design; SQL Analytical Investigation; Data Forensics Investigation; Final Report.

3.2 Phase 1 — Business Understanding

Purpose: Understand why the business needs analysis before touching the data. Work: The dataset was framed as a multi-channel retail company called DataCo Global, operating across international markets. Key areas identified: sales performance, profitability, customer segments, product/category performance, shipping/delivery, regional/market performance. A Business Requirement Document (BRD) was produced. Output: DataCo_BRD.pdf.

3.3 Phase 2 — Source System Discovery

Purpose: Understand the source systems that produced the data. Work: Two source datasets identified — DataCoSupplyChainDataset.csv (~180,519 rows, 53 columns, transactional order records) and tokenized_access_logs.csv (~415,000 rows, 8 columns, e-commerce clickstream). Output: DataCo_Source_System_Discovery.pdf. Key conclusions: System A is the primary transactional source; System B is supplementary clickstream; both are flat-file exports.

3.4 Phase 3 — Data Dictionary

Purpose: Define every important column so downstream work uses consistent business definitions. Work: All 61 fields across the two datasets documented (column name, business definition, data type, domain classification, example values). Fields grouped into seven domains: Customer, Product, Order Transaction, Shipping & Logistics, Financial Metrics, Geospatial, Clickstream. Output: DataCo_Data_Dictionary.pdf.

3.5 Phase 4 — Data Profiling

Purpose: Investigate actual contents rather than trusting documentation. Work: Domain-by-domain profiling of row/column counts, data types, missing values, duplicates, unique counts, distributions, invalid values, consistency. Key observations: Product Description 100% null; Order Zipcode ~86% null; 0 exact duplicate rows; 114,767 duplicate Order Ids (expected — grain is order-line item). Output: profiling.ipynb.

3.6 Phase 5 — Data Cleaning

Purpose: Produce an analytics-ready dataset without changing underlying business meaning. Work: Data type correction (dates); PII removal (email, name, password, street); low-value column removal; text standardization; feature engineering (year, month, quarter, weekday); business-rule validation (quantity > 0, price ≥ 0); deduplication; column name standardization. Final shape: 180,519 rows × 51 columns. Output: orders_clean.csv, cleaning.ipynb.

3.7 Phase 6 — Business System & Data Modeling

Purpose: Translate business reality into a structured data model. Work: Modeling began from the business system. Grain decision: one row = one order line item (smallest meaningful business event). Star schema: dim_customer, dim_date, fact_sales, dim_product. Shipping and location attributes remained in the fact table because they belong to individual business events. Output: 00_Mental_Model_of_Data_Modeling.md,

01_business_model.md.

3.8 Phase 7 — PostgreSQL Analytical Warehouse

Purpose: Move from a runtime dataset into a persistent analytical system. Work: Database dataco_supply_chain created with analytics schema. Four tables implemented: dim_customer, dim_product, dim_date, fact_sales. Landing table analytics.orders_clean loaded first; dimensions populated with DISTINCT selects and ON CONFLICT DO NOTHING; fact table populated preserving order_item_id as PK. Output: 01–06 SQL scripts.

3.9 Phase 8 — Warehouse Validation

Purpose: Prove the warehouse model did not silently change the data. Work: 20-test validation suite covering Completeness (landing = fact = 180,519), Uniqueness, Referential integrity (no orphans), Accuracy (sales/profit/quantity/discount reconciliation), Business rules (no null FKs, date range consistent). All 20 tests passed. Output: 07_warehouse_validation.sql. Status: READY FOR SQL ANALYTICS.

3.10 Phase 9 — Business Question Design

Purpose: Define business questions before writing SQL. Work: Business Question Map with 24 questions across eight domains: Sales Performance (Q1–Q4), Profitability (Q5–Q8), Product Performance (Q9–Q11), Customer Performance (Q12–Q14), Shipping & Delivery (Q15–Q18), Discount & Profitability (Q19–Q21), Market & Regional (Q22–Q23), Deep-Dive (Q24). Output: 07_00_Business_Question_Map.md.

3.11 Phase 10 — SQL Analytical Investigation

Purpose: Use the warehouse to investigate the business questions. Work: Eight structured investigations, each with Question, Grain & Source, Findings (Layer 1), Forensics Review (Layer 2), and Verdict. Investigations presented in full in Sections 4–11.

3.12 Phase 11 — Data Forensics Investigation

Purpose: Assess whether findings from Phase 10 can be trusted as evidence about a real business. Work: Nine findings confirmed (F1–F9). Each follows Claim / Evidence / Interpretation / Limitation structure. Presented in full in Section 12.

3.13 Phase 12 — Final Report

Purpose: Assemble all phases into a single, coherent, honest report. This document is the Phase 12 deliverable.

End of Section 3. Proceed to Section 4: Investigation 01.

SECTION 4 — INVESTIGATION 01: OVERALL PERFORMANCE

4.1 Question

What does the business represented by this dataset look like overall? What are the baseline sales, profit, order volume, and profitability figures?

4.2 Grain & Source

Grain: Order-line item. Source: fact_sales. Period: 2015-01-01 to 2017-09-30.

4.3 Baseline Metrics

Metric	Value
Unique orders	65,752
Order-line items	180,519
Average lines per order	2.75
Total sales	\$36,784,734.31
Total profit	\$3,966,902.97
Overall profit margin	10.79%
Average order value	\$559.45
Median order value	\$504.89

The difference between unique orders (65,752) and order-line items (180,519) confirms that the fact table grain is order-line item, not order. Every aggregate at line level must be interpreted at that grain.

4.4 Order Profitability Structure

Profitability	Orders	Sales	Profit
Profitable	51,722	\$28,669,786.94	+\$6,582,106.76
Loss-making	13,908	\$8,088,899.60	-\$2,615,203.79
Break-even	122	\$26,047.77	\$0.00
Total	65,752	\$36,784,734.31	\$3,966,902.97

Approximately 21.15% of orders are loss-making. These orders represent 22.0% of sales and consume 39.7% of the gross profit generated by profitable orders.

4.5 Loss Depth

Metric	Value
Median loss margin	-22.60%
Average loss margin	-37.78%
Worst single order margin	-269.50%
Median loss per order	-\$108.79
Average loss per order	-\$188.04
Deepest single loss	-\$4,325.85

The typical loss-making order loses ~\$109 (22.6% of sales). The average is worse than the median, indicating a tail of more extreme losses.

4.6 Order Value Distribution

Percentile	Value
Minimum	\$9.99
P25	\$265.96
Median	\$504.89
Average	\$559.45
P75	\$799.95
P95	\$1,199.94
P99	\$1,500.00
Maximum	\$3,449.91

Average (\$559) and median (\$505) differ by only ~10%. The distribution is mildly right-skewed. There is no "whale" concentration. The \$1,500 value at P99 is a hard anchor and was flagged for later investigation.

4.7 Hypotheses Eliminated

Hypothesis	Verdict
Discount caused losses	Eliminated (10.2% vs 10.2%)
Quantity caused losses	Eliminated (2.13 both groups)
Order status caused losses	Eliminated (all statuses positive)
Category caused losses	Eliminated (all profitable)
Product caused losses	Eliminated (only 3 tiny negatives)

4.8 Forensics Review

The baseline figures hold descriptively. However, a schema issue was flagged: `order_profit_per_order` appears identical to `benefit_per_order` on every row. Raw negative lines show impossible profit ratios (–1.5 to –2.55). Loss exceeds sale price in several cases. Hypothesis: the field may be an order-level measure repeated on every line item, which would inflate line-level profit aggregation. This is addressed in Forensics Finding F6. The \$1,500 value at P99 is flagged for later investigation (Forensics Finding F7).

4.9 Verdict

The business shows a 10.79% margin at the aggregate level. Approximately one in five orders is loss-making, with losses deep enough to consume a significant portion of profitable orders' contribution. The overall average is stable but the underlying order-level distribution is not.

End of Investigation 01. Proceed to Section 5: Investigation 02.

SECTION 5 — INVESTIGATION 02: TREND OVER TIME

5.1 Question

How do sales and profit move over time? Do they track each other, or diverge? Are there strong or weak periods?

5.2 Grain & Source

Grain: Month, Year. Source: fact_sales, dim_date. Period: 2015-01-01 to 2018-01-31 (raw); Valid: 2015-01-01 to 2017-09-30 (after data quality check).

5.3 Annual Performance

Year	Orders	Sales	Profit	Margin
2015	20,904	\$12,340,831	\$1,318,857	10.69%
2016	20,859	\$12,303,817	\$1,310,119	10.65%
2017*	15,586	\$9,603,617	\$1,057,009	11.01%

*2017 covers January–September only. Annualized figures are approximately 4–7% higher than 2016.

5.4 Findings

- 2015 and 2016 are statistically identical. The business was flat in that period.
- 2017 shows slight growth in sales (+4%) and profit (+7%) when annualized, with essentially the same order count.
- Margin is stable at approximately 10.7% – 11.0%.
- No deterioration pattern detected.

5.5 Structural Break

A structural break was identified at October 2017: average lines per order drops from ~3.0 to exactly 1.0; daily order count alternates between 68 and 69; sales values repeat across days; sales per order drops from ~\$660 to ~\$156.

5.6 Forensics Review

The final four months of the dataset (October 2017 – January 2018) are synthetic / placeholder data. The pattern is not consistent with real business activity. This is Forensics Finding F2. The dataset's valid analysis window is therefore 2015-01-01 to 2017-09-30.

5.7 Verdict

Within the valid analysis window, the business is mature and stable: flat 2015–2016, slight growth in 2017, with margin holding steady at ~10.7–11.0%. The apparent collapse at the end of 2017 is a data-generation artifact, not a business event.

End of Investigation 02. Proceed to Section 6: Investigation 03.

SECTION 6 — INVESTIGATION 03: MARKETS & REGIONS

6.1 Question

Which markets and regions generate the most sales and profit? Are there high-sales but low-profit markets? Are there geographic profitability differences?

6.2 Grain & Source

Grain: Market, Region (whole period). Source: fact_sales. Period: 2015-01-01 to 2017-09-30.

6.3 Market Performance

Market	Orders	Sales	Profit	Margin
LATAM	17,181	\$10,277,613	\$1,123,322	10.93%
Europe	15,778	\$9,596,779	\$1,036,634	10.80%
Pacific Asia	11,957	\$7,012,892	\$709,644	10.12%
USCA	8,579	\$5,066,529	\$564,314	11.14%
Africa	3,854	\$2,294,453	\$252,071	10.99%

Sales range: \$2.29M – \$10.28M (4.5× spread). Margin range: 10.12% – 11.14% (1.02 percentage points). LATAM leads by sales. Africa is smallest. No market is broken.

6.4 Region Performance (top 5 by sales)

Region	Orders	Sales	Margin
Central America	9,396	\$5,665,712	10.88%
Western Europe	8,395	\$5,152,093	10.70%
South America	4,979	\$2,960,881	11.32%
Northern Europe	3,114	\$1,878,913	10.88%
Southern Europe	2,977	\$1,791,507	11.25%

Regional margin range: 9.64% – 13.51%.

6.5 Findings

- Sales vary significantly by market.
- Margins are remarkably uniform across markets and regions.
- No market or region is loss-making.
- The business is geographically unbalanced by volume but uniform by margin.

6.6 Data Quality Issue

Market labels are not stable across years. Entire market-years are missing from the data:

Market	2015	2016	2017
Africa	0	3,445	409
Europe	8,315	1,274	6,189
LATAM	8,598	0	8,583
Pacific Asia	3,991	7,602	364
USCA	0	8,538	41

Growth-by-market analysis is therefore not reliable.

6.7 Forensics Review

Market and region analysis is reliable at the whole-period level but not year-over-year. Market labels appear to have been generated or reassigned inconsistently (Forensics Finding F3). The uniform margins across all markets contribute to Forensics Finding F4.

6.8 Verdict

The business has real volume concentration (LATAM + Europe $\approx$ 54% of sales) but almost no margin differentiation. No region requires strategic intervention based on margin.

End of Investigation 03. Proceed to Section 7: Investigation 04.

SECTION 7 — INVESTIGATION 04: PRODUCTS & CATEGORIES

7.1 Question

Which products and categories generate the most sales and profit? Are there high-sales but weak-profit products? Are there loss-making categories?

7.2 Grain & Source

Grain: Category, then Product. Source: fact_sales, dim_product. Period: 2015-01-01 to 2017-09-30.

7.3 Category-Level Performance

Most categories show approximately 10–11% margin. Big-volume categories are uniform at approximately 10–11%. Three small anomalies:

Category	Margin	Sales
Basketball	–2.64%	\$19K
Strength Training	1.55%	\$37K
As Seen on TV!	3.47%	\$20K

Top categories by sales:

Category	Sales	Margin
Fishing	\$6,922,854	10.90%
Cleats	\$4,424,744	11.16%
Camping & Hiking	\$4,113,926	10.37%
Cardio Equipment	\$3,688,444	10.35%
Women's Apparel	\$3,143,900	11.14%

7.4 Product-Level Performance

Top 30 products by sales show margins between 6.97% and 14.58%. No high-sales product is loss-making at scale.

7.5 Data Quality Issue

Product names do not match their assigned categories:

Product	Category
Field & Stream Sportsman 16 Gun Fire Safe	Fishing
Perfect Fitness Perfect Rip Deck	Cleats
Nike Men's Free 5.0+ Running Shoe	Cardio Equipment
Titleist Pro V1x Golf Balls	Electronics
O'Brien Men's Neoprene Life Vest	Indoor/Outdoor Games

Category sizes are also suspiciously balanced (\$3–7M each).

7.6 Forensics Review

Category-level analysis is **NOT TRUSTWORTHY**. Categories appear to be randomly assigned. This is Forensics Finding F5. What CAN be said: the dataset assigns products to 50 categories, but the assignment appears

random. Category-level conclusions are invalid. What CANNOT be said: "Category X is the most profitable"; "Category Y underperforms"; any category-level business insight.

7.7 Verdict

The dataset's category dimension is not reliable. Category performance cannot be interpreted as evidence about real product groupings.

End of Investigation 04. Proceed to Section 8: Investigation 05.

SECTION 8 — INVESTIGATION 05: CUSTOMER PERFORMANCE

8.1 Question

Which customer segments generate the most sales and profit? Are high-value segments also profitable? Do any segments show notably different economics?

8.2 Grain & Source

Grain: Customer segment. Source: fact_sales, dim_customer. Period: 2015-01-01 to 2017-09-30.

8.3 Segment-Level Performance

Segment	Orders	Customers	Sales	Profit	Margin
Consumer	29,789	6,409	\$17,766,498	\$1,915,896	10.78%
Corporate	17,304	3,709	\$10,386,280	\$1,118,095	10.77%
Home Office	10,256	2,212	\$6,095,487	\$651,995	10.70%

8.4 Sales per Customer

Segment	Sales per Customer
Consumer	\$2,772
Corporate	\$2,800
Home Office	\$2,756

8.5 Findings

- Margins are nearly identical across all three segments.
- Sales per customer is nearly identical (\$2,756 – \$2,800).
- Consumer drives the highest volume by a large margin.
- No segment is broken. No segment is a "danger" segment.

8.6 Uniformity Pattern Confirmation

This is the fourth dimension showing uniform margins:

Dimension	Margin Range
Markets	10.12% – 11.14%
Regions	9.64% – 13.51%
Categories	9.90% – 17.46%
Segments	10.70% – 10.78% (tightest yet)

8.7 Forensics Review

Customer segment analysis is **NOT TRUSTWORTHY**. The segment field does not produce meaningfully different business behavior. This is consistent with the dataset being synthetic or partially randomized (Forensics Findings F4 and F5). What CAN be said: in this dataset, three customer segments exist, but they show nearly identical profitability. What CANNOT be said: "Consumer customers are more profitable than Corporate"; "Focus on the Corporate segment"; any segment-level business recommendation.

8.8 Verdict

The customer segment dimension is present in the dataset but does not separate business behavior. Segment-level analysis cannot be used to support strategy.

End of Investigation 05. Proceed to Section 9: Investigation 06.

SECTION 9 — INVESTIGATION 06: SHIPPING PERFORMANCE

9.1 Question

What is the overall late-delivery rate? Does late risk differ by shipping mode? How do actual vs scheduled shipping times compare?

9.2 Grain & Source

Grain: Order (deduplicated from line-level fact). Source: fact_sales. Period: 2015-01-01 to 2017-09-30.

9.3 Shipping Mode Performance

Shipping Mode	Orders	Late Risk	Real Days	Scheduled	Gap
Standard Class	34,331	38.14%	4.00	4.00	0.00
Second Class	11,128	76.68%	4.00	2.00	2.00
First Class	8,786	95.25%	2.00	1.00	1.00
Same Day	3,104	45.65%	0.48	0.00	0.48

9.4 Findings

- Shipping is the first dimension showing real variance in the data (late risk ranges from 38% to 95%).
- The pattern is backwards. Standard Class (cheapest) shows the lowest late risk; First Class (most expensive) shows the highest.
- Shipping days are round numbers (4.00, 2.00, 1.00), with no within-mode variance.
- Standard Class shows zero day-variance across all 34,331 orders.

9.5 Data Quality Issue

Standard Class shipping has an average real shipping time of exactly 4.00 days and a scheduled time of exactly 4.00 days across all 34,331 orders. Real operations always exhibit variance. Zero variance is a strong signal of synthetic construction.

9.6 Forensics Review

Shipping data is **SYNTHETIC**. The observed variance (38–95% late risk) is real in the data, but the underlying patterns do not reflect plausible logistics behavior. This is Forensics Finding F9. Additional confirmation: late_delivery_risk is constant within each order (order-level field repeated on lines), consistent with Forensics Finding F6. What CAN be said: in this dataset, shipping modes have different recorded late rates (38–95%), but the underlying timing patterns are implausible. What CANNOT be said: "First Class shipping is unreliable"; "Standard Class is the most reliable mode"; any shipping-mode business recommendation.

9.7 Verdict

Shipping data cannot be interpreted as evidence about real logistics performance. The variance is an artifact of the dataset's construction.

End of Investigation 06. Proceed to Section 10: Investigation 07.

SECTION 10 — INVESTIGATION 07: DISCOUNT & PROFITABILITY

10.1 Question

What is the average discount rate? How does discount vary across lines? Is discount associated with lower profit?

10.2 Grain & Source

Grain: Line item. Source: fact_sales. Period: 2015-01-01 to 2017-09-30.

10.3 Discount Statistics

Statistic	Value
Total lines	171,962
Average discount	10.17%
Minimum	0.00%
P25	4.00%
Median	10.00%
P75	16.00%
Maximum	25.00%

10.4 Discount Distribution

Every distinct discount rate has approximately 9,550 lines. Rates present: 0–7%, 9–10%, 12–13%, 15–18%, 20%, 25% (18 distinct values). Rates missing: 8%, 11%, 14%, 19%, 21–24%. This equal-frequency distribution is not consistent with real discount practice. Real discount systems cluster at strategic values (10%, 15%, 20%) and exhibit uneven frequency.

10.5 Effect of Discount on Profit

Discount	Lines	Avg Profit Ratio	Avg Line Profit
0%	9,550	0.1256	\$25.67
10%	9,554	0.1209	\$22.05
25%	9,551	0.1271	\$19.12

10.6 Findings

- The discount rate distribution is artificially equalized.
- Profit ratio is flat across all discount rates (12–13%).
- Profit dollars drop with discount, but only mechanically (the arithmetic effect of discount on sales).
- No behavioral signal exists.

10.7 Forensics Review

Discount data is **SYNTHETIC**. The discount field was generated to appear varied, but the distribution is mathematically uniform. This is Forensics Finding F8. What CAN be said: in this dataset, discount rates are evenly distributed across 18 values, and profit ratio does not vary with discount. What CANNOT be said: "Discounts reduce profitability" (mechanical artifact); "Product X is heavily discounted" (random distribution); any discount-related business recommendation.

10.8 Verdict

Discount data does not reflect real pricing behavior. The distribution is an artifact of construction.

End of Investigation 07. Proceed to Section 11: Investigation 08.

SECTION 11 — INVESTIGATION 08: CROSS-DIMENSION DEEP DIVE

11.1 Question

Do any cross-dimension combinations reveal real patterns that single dimensions did not? Are loss-making lines concentrated in specific combinations?

11.2 Grain & Source

Grain: Multi-dimensional (market × shipping_mode as test case). Source: fact_sales. Period: 2015-01-01 to 2017-09-30.

11.3 Market × Shipping Mode Combination

Selected combinations ordered by average line profit:

Market	Shipping Mode	Lines	Avg Profit	Margin
Africa	Same Day	668	\$17.48	8.68%
Pacific Asia	Second Class	7,035	\$18.32	9.34%
Africa	Second Class	2,164	\$18.65	9.49%
Pacific Asia	Same Day	1,908	\$19.46	10.07%
Pacific Asia	Standard Class	21,247	\$20.36	10.33%
Africa	First Class	1,727	\$24.78	12.71%

11.4 Findings

- Margin range: 8.68% – 12.71% (4.03 percentage points).
- Extreme values appear only in small combinations.
- Large combinations (>20,000 lines) converge to ~10.8%.
- This is the signature of small-sample noise, not business signal.

11.5 Forensics Review

No cross-dimension pattern exists. The dataset is uniformly synthetic at every dimensional level and every combination. This confirms the collective forensics findings F2–F9. What CAN be said: in this dataset, all cross-dimension combinations show margins clustering around ~10.8%, with deviations explained by small-sample noise. What CANNOT be said: "Africa + Same Day is a loss-making combination"; any cross-dimension business insight.

11.6 Verdict

Cross-dimension analysis produces no new signal. Every dimension and every combination reflects the same uniform construction already identified in the forensics findings.

End of Investigation 08. Proceed to Section 12: Data Forensics Findings.

SECTION 12 — DATA FORENSICS FINDINGS

12.1 Purpose

This section consolidates every data quality and trust issue uncovered during the investigation (Investigations 01–08). Each finding follows a standard structure: CLAIM — what was observed; EVIDENCE — the data that supports it; INTERPRETATION — what it means; LIMITATION — what cannot be confirmed; STATUS — confirmed / open. The goal is not to attack the dataset. The goal is to determine what the dataset CAN and CANNOT support as evidence. The chapter is the centerpiece of the project.

12.2 F1 — The Dataset Is a Research Artifact, Not Real Business Data

CLAIM

The DataCo Supply Chain dataset is a research artifact, not recorded transactions from a real company.

EVIDENCE

- Kaggle source: "DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS"
- Author: Shashwat Tiwari, Instituto Politecnico de Leiria
- Purpose: Big Data / Machine Learning research; Tagged: Data Visualization, EDA, Research

INTERPRETATION

The "DataCo business" is a construct for research purposes. Business findings derived from this dataset do not describe a real company.

LIMITATION

Some datasets are semi-synthetic, combining real patterns with synthetic additions. We cannot rule out that portions of this dataset reflect real supply chain structure.

STATUS: Confirmed.

12.3 F2 — October 2017 Onward Is Synthetic

CLAIM

The last four months of the dataset (October 2017 – January 2018) are synthetic or placeholder data.

EVIDENCE

- Average lines per order drops from ~3.0 to exactly 1.0
- Daily order count alternates between 68 and 69 for every day
- Sales values repeat across days; Sales per order drops from ~\$660 to ~\$156

INTERPRETATION

This is not a business collapse. It is a data generation artifact — a synthetic fill at the end of the dataset.

LIMITATION

The reason for this fill cannot be determined from the dataset alone. It may be intentional or accidental.

IMPACT: All trend analysis must EXCLUDE October 2017 onward. Valid window: 2015-01-01 to 2017-09-30.

STATUS: Confirmed.

12.4 F3 — Market Labels Are Unstable Across Years

CLAIM

The market field in fact_sales is not consistent year-to-year. Entire market-years are missing.

EVIDENCE

Market-year order counts show Africa: 0/3,445/409; Europe: 8,315/1,274/6,189; LATAM: 8,598/0/8,583; Pacific Asia: 3,991/7,602/364; USCA: 0/8,538/41. Annual totals still add up; no rows lost in aggregate.

INTERPRETATION

Market assignments were not maintained consistently across years. Either the dataset was assembled from multiple sources, or market labels were generated for research purposes.

LIMITATION

Market-level analysis is only reliable at the whole-period level. Growth-by-market cannot be analyzed.

STATUS: Confirmed.

12.5 F4 — Margins Are Uniformly ~10–11% Across Every Dimension

CLAIM

Every major dimension (market, region, category, segment, shipping mode) shows margins that are suspiciously uniform.

EVIDENCE

Market margins: 10.12%–11.14% (1.02 pp); Region: 9.64%–13.51% (3.87 pp); Category: 9.90%–17.46% (7.56 pp, excl. anomalies); Segment: 10.70%–10.78% (0.08 pp).

INTERPRETATION

Real retail businesses do NOT have uniform margins across markets, regions, categories, and segments. This level of uniformity is a signature of data construction — a fixed margin rule applied across the dataset.

LIMITATION

It is theoretically possible (though very unusual) for a real business to operate with a uniform pricing and cost model. Combined with other findings, synthetic construction is the more parsimonious explanation.

STATUS: Confirmed.

12.6 F5 — Categories Are Randomly Assigned

CLAIM

Product categories in `dim_product` do not reflect real product groupings. Category assignments appear shuffled.

EVIDENCE

"Field & Stream Sportsman 16 Gun Fire Safe" → "Fishing"; "Perfect Fitness Perfect Rip Deck" → "Cleats"; "Nike Men's Free 5.0+ Running Shoe" → "Cardio Equipment"; "Titleist Pro V1x Golf Balls" → "Electronics"; "O'Brien Men's Neoprene Life Vest" → "Indoor/Outdoor Games". Category sizes suspiciously balanced (\$3–7M each).

INTERPRETATION

The category field is not reliable as a business dimension. Category-level analysis is not meaningful.

LIMITATION

Some categories may still be internally consistent at the product level. The shuffling may be at the category-name level rather than per-product.

IMPACT: Category should NOT appear as a business dimension in any downstream analysis or report.

STATUS: Confirmed.

12.7 F6 — `order_profit_per_order` May Be Order-Level, Not Line-Level

CLAIM

`order_profit_per_order` may be an order-level measure repeated on every line item, causing inflation when summed at line level.

EVIDENCE

`order_profit_per_order` = `benefit_per_order` (identical on every row); raw negative lines show impossible ratios (–1.5 to –2.55); loss exceeds sale price (e.g., –\$4,274.98 on a \$1,999.99 sale); `late_delivery_risk` is constant within each order.

INTERPRETATION

If the field is order-level repeated on lines, then total profit at line level is INFLATED; the "21% loss-making orders" finding may be overstated; the "\$2.62M total loss" may be smaller than reported.

LIMITATION

We have not confirmed this with a direct deduplication test. The field could also be a line-level measure that happens to be equal across lines in many orders.

STATUS: Open. Requires verification.

12.8 F7 — The \$1,500 Price Anchor

CLAIM

\$1,500 appears as a suspicious recurring value — both as an order-value ceiling and as a product price in the most extreme negative-margin lines.

EVIDENCE

P99 order value = exactly \$1,500.00; top negative lines cluster around \$1,500 product price; P95 order value = \$1,199.94.

INTERPRETATION

\$1,500 is likely a hard ceiling in the data, or a common pricing point. It could be intentional or a data construction artifact.

LIMITATION

We have not investigated whether \$1,500 products behave consistently or whether the price is synthetic.

STATUS: Open. Could be investigated in future work.

12.9 F8 — Discount Distribution Is Artificial

CLAIM

Discount rates are artificially equalized across 18 distinct values.

EVIDENCE

All 18 distinct discount rates appear approximately 9,550 times each; missing rates: 8%, 11%, 14%, 19%, 21–24%; maximum discount = 25% (hard cap); profit ratio is flat across all discount rates (~12–13%).

INTERPRETATION

The discount field was generated to appear varied, but the distribution is mathematically uniform. This is not consistent with any plausible real discount strategy.

LIMITATION

It is theoretically possible for a real business to enforce a limited set of discount values, but the equal-frequency property cannot arise from real pricing behavior.

IMPACT: Discount strategy analysis is invalid on this dataset.

STATUS: Confirmed.

12.10 F9 — Shipping Data Is Synthetic

CLAIM

Shipping data has zero variance within modes and exhibits backwards patterns.

EVIDENCE

Standard Class: real days = scheduled days = 4.00 across all 34,331 orders (gap = 0.00); First Class: 95.25% late risk vs Standard Class 38.14% (backwards from expected logistics); Same Day: averages 0.48 days (not same-day); all day values are round numbers.

INTERPRETATION

The variance observed (38–95% late risk) is real in the data, but the underlying patterns do not reflect plausible logistics behavior.

LIMITATION

We did not directly inspect carrier-level data (not present in the dataset). Conclusions rest on internal consistency of the shipping fields.

IMPACT: Shipping performance analysis is invalid on this dataset.

STATUS: Confirmed.

12.11 Summary Table

ID	Finding	Status
F1	Dataset is a research artifact	Confirmed
F2	October 2017 onward is synthetic	Confirmed
F3	Market labels unstable across years	Confirmed
F4	Margins uniformly ~10–11% across dimensions	Confirmed
F5	Categories are randomly assigned	Confirmed
F6	order_profit_per_order may be order-level	Open
F7	\$1,500 price anchor	Open
F8	Discount distribution is artificial	Confirmed
F9	Shipping data is synthetic	Confirmed

12.12 What This Means for the Project

The dataset cannot support real business insights. It CAN support: practicing the full analyst workflow; data quality assessment and forensics demonstration; methodological demonstration; machine learning experimentation on structured data. It CANNOT support: real business strategy recommendations; market, regional, or category insights; profitability analysis (given F6 uncertainty); discount strategy analysis (given F8); shipping performance analysis (given F9).

12.13 Recommendations for Future Users of This Dataset

If you are using the DataCo dataset for analysis:

1. Exclude October 2017 – January 2018 (synthetic data).
2. Do not trust category assignments.
3. Do not do market growth analysis (unstable labels).
4. Verify whether order_profit_per_order is order-level or line-level before using it.
5. Treat any "uniform pattern" finding with suspicion — it may be a generation artifact.

If using for machine learning: the dataset remains useful for training models on structured/tabular data, practicing EDA workflows, and benchmarking algorithms — but do not treat learned patterns as reflecting real business dynamics.

End of Section 12. Proceed to Section 13: What This Dataset Can Support.

SECTION 13 — WHAT THIS DATASET CAN SUPPORT

13.1 Purpose

This section summarizes the practical uses of the DataCo dataset that remain valid after the forensics findings in Section 12. The dataset is not worthless simply because it is synthetic. It retains real value for specific use cases.

13.2 Methodological Practice

The dataset supports full end-to-end analytics practice: business understanding and requirement documentation; source system discovery; data dictionary construction; data profiling and quality assessment; data cleaning with business-meaning preservation; star schema design and justification; PostgreSQL warehouse construction; warehouse validation (20-test suite); business question design before SQL; structured SQL investigation; data forensics assessment; honest analytical reporting. The dataset's size (~180,000 rows) and richness (51 columns, multiple dimensions) make it suitable for building a realistic analytical workflow.

13.3 Data Forensics Demonstration

The dataset is an excellent case for data forensics practice and education: detecting synthetic patterns in structured data; detecting randomized categorical assignments; detecting uniform margins across dimensions; detecting schema issues (order-level fields repeated on lines); documenting findings with evidence and limitations; reporting honestly about what data can and cannot support. Real datasets with clean integrity do not provide these opportunities.

13.4 Machine Learning Experimentation

The dataset remains useful for machine learning work: training models on structured/tabular data; practicing feature engineering; benchmarking classification and regression algorithms; testing preprocessing pipelines; evaluating cross-validation strategies. Limitation: learned patterns should not be interpreted as reflecting real business dynamics.

13.5 Education and Coursework

The dataset is suitable for analytics and supply chain courses: teaching relational modeling and star schema design; teaching SQL joins, aggregates, window functions; teaching data quality concepts; teaching investigative methodology; teaching the difference between descriptive findings and business insights. The dataset's quirks create useful teaching moments.

13.6 Reproducibility and Benchmarking

The dataset's stable, synthetic nature makes it a useful control case: reproducible analytical workflows; comparing analytical tools (PostgreSQL vs. pandas vs. other engines); benchmarking data quality tools; testing dashboards and BI systems under known conditions. Because the dataset does not change, results are comparable across tools and versions.

13.7 Summary

The dataset CAN support: full analyst workflow practice; data forensics methodology; machine learning experimentation; education and coursework; reproducibility and benchmarking. What the dataset CANNOT support is real business insight. That distinction is documented in Section 14.

End of Section 13. Proceed to Section 14: What This Dataset Cannot Support.

SECTION 14 — WHAT THIS DATASET CANNOT SUPPORT

14.1 Purpose

This section explicitly documents the analytical uses that the DataCo dataset does NOT support. The goal is to prevent future users from drawing unsupported conclusions. Each limitation corresponds to one or more of the forensics findings in Section 12.

14.2 Real Business Insights

Cannot say any statement framed as insight about a real DataCo company. Why: Forensics Finding F1 — the dataset is a research artifact.

14.3 Strategic Recommendations

Cannot say "DataCo should invest more in category X," expand into market Y, reduce discounting in region Z, or any recommendation that assumes the dataset describes real business performance. Why: F3 (market labels unstable), F5 (categories randomly assigned), combined with F1.

14.4 Market and Regional Analysis

Cannot say "Region X underperforms Region Y," "Market growth is driven by region Z," or similar. Why: F3 (labels unstable across years); F4 (margins uniformly ~10–11%).

14.5 Category and Product Grouping Analysis

Cannot say "Category X is the most profitable" or any category-level assortment decision. Why: F5 — categories are randomly assigned; product names do not match assigned categories.

14.6 Customer Segment Analysis

Cannot say "Consumer segment is more profitable than Corporate" or focus strategy on a segment. Why: F4 — segment margins are indistinguishable (10.70%–10.78%).

14.7 Profitability Analysis

Cannot say "The business generates X% margin on its portfolio" or product-level profit conclusions that depend on line-level aggregation. Why: F6 — `order_profit_per_order` appears to be an order-level field repeated on every line item.

14.8 Growth Analysis by Market

Cannot say "Market X grew by Y% year-over-year." Why: F3 — market labels unstable; entire market-years missing.

14.9 Discount Strategy Analysis

Cannot say "Discounts reduce profitability" or "Product X is heavily discounted." Why: F8 — discount distribution artificially equalized; profit ratio flat across rates.

14.10 Shipping Performance Analysis

Cannot say "First Class shipping is unreliable" or "Standard Class is the most reliable mode." Why: F9 — shipping data has zero within-mode variance and backwards patterns.

14.11 Causal Analysis

Cannot say "X caused Y" or any statement establishing one variable as a genuine driver of another. Why: All findings in Section 12 combine to rule out causal interpretation.

14.12 Summary Table

Analytical use	Supported?
Real business insights	No
Strategic recommendations	No
Market analysis	No
Category / product grouping	No
Customer segment analysis	No
Profitability analysis	No
Growth-by-market analysis	No
Discount strategy analysis	No
Shipping performance analysis	No
Causal analysis	No
Analyst workflow practice	Yes
Data forensics demonstration	Yes
Machine learning experimentation	Yes
Education and coursework	Yes
Reproducibility and benchmarking	Yes

End of Section 14. Proceed to Section 15: Conclusion.

SECTION 15 — CONCLUSION

15.1 Project Summary

This project set out to build an end-to-end supply chain analytics system on the DataCo dataset and answer real business questions about sales, profitability, customers, products, shipping, and geography. It delivered a full analytics build: business understanding and documentation; source system discovery; data dictionary; data profiling; data cleaning; star schema modeling; PostgreSQL warehouse construction; 20-test warehouse validation; business question design (24 questions across 8 domains); eight structured SQL investigations; data forensics assessment (9 findings); and this final report.

15.2 Central Finding

During the investigations, evidence emerged that the dataset is partially synthetic. The forensics work confirmed nine distinct findings: the dataset is a research artifact, not a real company; a large portion (October 2017 – January 2018) is fully synthetic; market labels are inconsistent across years; margins are uniformly ~10–11% across every dimension; categories are randomly assigned; a schema issue (F6) means line-level profit aggregation is suspect; a \$1,500 price anchor appears; discount distribution is artificially equalized; shipping data has zero variance and backwards patterns. Taken together, these findings mean that the dataset cannot be used to generate real business insights. The "DataCo business" is a research construct.

15.3 What the Project Demonstrates

The project demonstrates the full analyst workflow: business understanding; source system discovery; data dictionary construction; data profiling; data cleaning with business-meaning preservation; star schema design and justification; PostgreSQL warehouse construction; warehouse validation; business question design before SQL; structured SQL investigation; data forensics assessment; honest analytical reporting; clear communication of what data can and cannot support. Every phase was executed and documented. Every investigation was executed and reported. Every forensics finding was documented with evidence and limitations.

15.4 The Honest Conclusion

The deliverable is not a business analysis of a fictional company. It is a rigorous audit of a research dataset, documenting its limitations with evidence. This is a professional, evidence-based conclusion. Most analytical projects present findings without questioning whether the data deserves interpretation. This project does both. The dataset is not worthless. It retains real value for analyst workflow practice, data forensics demonstration, machine learning experimentation, education, and reproducibility testing. But it does not support business interpretation, and that distinction is documented honestly throughout this report.

15.5 Closing Note

A mature analyst does not assume that every dataset deserves interpretation. A mature analyst investigates, tests, and documents — and, when necessary, reports that the data cannot support real conclusions. That is what this project does.

End of Section 15. Proceed to Appendix A.

APPENDIX A — BUSINESS QUESTION MAP

A.1 Purpose

This appendix documents the business questions that the DataCo analytical warehouse was designed to answer. The questions are derived from the business system, available data, and the star schema model created in Phase 6. The goal was not to write SQL randomly. The goal was to establish: Business Goal → Business Decision → Business Question → Metric → Data → SQL → Insight.

A.2 Domain 1 — Sales Performance

Q1. What is the total sales generated?

Metric: Total Sales | Data: fact_sales.sales

Q2. How do sales change over time?

Metric: Sales by month / year | Data: fact_sales.sales, dim_date

Q3. Which markets generate the most sales?

Metric: Sales by market | Data: fact_sales.sales, fact_sales.market

Q4. Which regions generate the most sales?

Metric: Sales by region | Data: fact_sales.sales, fact_sales.order_region

A.3 Domain 2 — Profitability

Q5. What is the total profit generated?

Metric: Total Profit | Data: fact_sales.order_profit_per_order

Q6. Which products generate the most profit?

Metric: Profit by product | Data: fact_sales.order_profit_per_order, dim_product.product_name

Q7. Which regions generate the most profit?

Metric: Profit by region | Data: fact_sales.order_profit_per_order, fact_sales.order_region

Q8. Does high sales performance correspond to high profit?

Metric: Sales vs Profit | Data: fact_sales.sales, fact_sales.order_profit_per_order

A.4 Domain 3 — Product Performance

Q9. Which products generate the highest sales?

Metric: Sales by product | Data: fact_sales.sales, dim_product.product_name

Q10. Which categories generate the highest sales and profit?

Metric: Sales and profit by category | Data: fact_sales, dim_product.category_name

Q11. Which products have high sales but weak profit?

Metric: Sales and profit | Data: fact_sales, dim_product.product_name

A.5 Domain 4 — Customer Performance

Q12. Which customer segments generate the most sales?

Metric: Sales by segment | Data: fact_sales.sales, dim_customer.customer_segment

Q13. Which customer segments generate the most profit?

Metric: Profit by segment | Data: fact_sales.order_profit_per_order, dim_customer.customer_segment

Q14. Which customers have the highest recorded sales and profit?

Metric: Sales and profit by customer | Data: fact_sales, dim_customer.customer_id

A.6 Domain 5 — Shipping & Delivery Performance

Q15. What proportion of recorded order-item rows have late-delivery risk?

Metric: Late Delivery Risk Rate | Data: fact_sales.late_delivery_risk

Q16. Which shipping modes have the highest late-delivery risk?

Metric: Late Risk by mode | Data: fact_sales.shipping_mode, fact_sales.late_delivery_risk

Q17. How does actual shipping time compare with scheduled?

Metric: Shipping Time Variance | Data: fact_sales.days_for_shipping_real, fact_sales.days_for_shipment_scheduled

Q18. Which regions experience the most delivery problems?

Metric: Late Risk Rate, Shipping Variance by region | Data: fact_sales.order_region, late_delivery_risk, days fields

A.7 Domain 6 — Discount & Profitability

Q19. How much discount is being given?

Metric: Total / Avg Discount, Discount Rate | Data: fact_sales.order_item_discount, order_item_discount_rate

Q20. Are higher discounts associated with lower profit?

Metric: Discount vs Profit | Data: discount fields, order_profit_per_order

Q21. Which products or categories rely heavily on discounts?

Metric: Average discount by product / category | Data: fact_sales, dim_product

A.8 Domain 7 — Market & Regional Investigation

Q22. Which markets have strong sales but weak profit?

Metric: Sales vs Profit by market | Data: fact_sales.market, sales, order_profit_per_order

Q23. Which regions have strong sales but weak profit?

Metric: Sales vs Profit by region | Data: fact_sales.order_region, sales, order_profit_per_order

A.9 Domain 8 — Deep-Dive Investigation

Q24. What factors appear alongside poor profitability?

Metric: Multi-dimensional | Data: fact_sales, dim_product, dim_customer, dim_date

A.10 Note on Application

The 24 questions above were used as the design input for the investigations presented in Sections 4–11. During the investigations, the Business Question Map was also used to guide which questions were worth investigating first, and which were not supported by the dataset's grain or reliability.

End of Appendix A. Proceed to Appendix B.

APPENDIX B — WAREHOUSE SCHEMA & VALIDATION

B.1 Warehouse Overview

Property	Value
Database	dataco_supply_chain
Schema	analytics
Model	Star schema
Grain	Order-line item (fact_sales)
Rows	180,519 (fact_sales)

B.2 Dimension Tables

B.2.1 dim_customer

customer_id (INTEGER PK), customer_segment, customer_city, customer_state, customer_country, customer_zipcode, latitude, longitude

B.2.2 dim_product

product_card_id (INTEGER PK), product_name, product_price, product_status, category_id, category_name, department_id, department_name

B.2.3 dim_date

order_date_dateorders (TIMESTAMP PK), order_year, order_month, order_quarter, order_weekday

B.3 Fact Table — fact_sales

order_item_id (INTEGER PK), order_id, customer_id (FK), product_card_id (FK), order_date_dateorders (FK), type, sales, order_profit_per_order, benefit_per_order, order_item_total, sales_per_customer, order_item_discount, order_item_discount_rate, order_item_quantity, order_item_product_price, order_item_profit_ratio, shipping_mode, delivery_status, late_delivery_risk, days_for_shipping_real, days_for_shipment_scheduled, shipping_date_dateorders, order_status, market, order_city, order_country, order_region, order_state, order_zipcode.

B.4 Referential Integrity

- fact_sales.customer_id → dim_customer.customer_id
- fact_sales.product_card_id → dim_product.product_card_id
- fact_sales.order_date_dateorders → dim_date.order_date_dateorders

B.5 Validation Suite

Validation was executed in 20 tests across five principles: Completeness (landing = fact = 180,519); Uniqueness (total fact rows = unique order_item_id; no duplicate dimension keys); Referential integrity (no orphan FKs); Accuracy (landing sales/profit/quantity/discount = fact); Business rules (no NULL FKs; date range identical). All 20 tests passed.

B.6 Warehouse Status

READY FOR SQL ANALYTICS.

End of Appendix B. Proceed to Appendix C.

APPENDIX C — SQL SCRIPTS (REFERENCE)

C.1 Overview

The SQL scripts that built and validated the warehouse, and that powered the eight investigations, are referenced here. The scripts are organized by phase.

C.2 Warehouse Construction Scripts

- 01_schema_setup.sql — CREATE SCHEMA IF NOT EXISTS analytics
- 02_check_landing_table.sql — Verify landing table loaded from Python/to_sql
- 03_create_dimensions.sql — Create dim_customer, dim_product, dim_date
- 04_populate_dimensions.sql — INSERT DISTINCT rows from orders_clean with ON CONFLICT DO NOTHING
- 05_create_fact_table.sql — Create fact_sales with FK constraints
- 06_populate_fact_table.sql — INSERT from orders_clean with ON CONFLICT DO NOTHING
- 07_warehouse_validation.sql — 20 validation tests across 5 principles

C.3 Investigation Scripts

Investigation queries for Investigations 01–08 are organized by investigation number: Investigation 01 — Baseline, order profitability, loss depth; Investigation 02 — Monthly trend, annual totals; Investigation 03 — Market and region performance; Investigation 04 — Category and product performance; Investigation 05 — Customer segment analysis; Investigation 06 — Shipping mode analysis; Investigation 07 — Discount distribution analysis; Investigation 08 — Cross-dimension analysis.

C.4 Accessing the Scripts

All SQL scripts are provided as .sql files alongside this report. They are standalone and can be executed against a PostgreSQL instance with the analytics schema and the orders_clean landing table loaded.

End of Appendix C. Proceed to Appendix D.

APPENDIX D — REFERENCE DATA

D.1 Dataset Source

Property	Value
Title	DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS
Source	Kaggle
Author	Shashwat Tiwari, Instituto Politecnico de Leiria
Purpose	Big Data / Machine Learning research
DOI	Mendeley Data — 8gx2fvg2k6 (version 5)
License	CC0: Public Domain

D.2 Analysis Window

2015-01-01 to 2017-09-30. Excluded period: October 2017 – January 2018 (synthetic).

D.3 Warehouse Environment

Database: PostgreSQL. Schema: analytics. Tooling: Python 3.13, pandas, numpy, SQLAlchemy, psycopg2-binary.

D.4 References to Project Artifacts

The following artifacts were produced during the project: DataCo_BRD.pdf (Phase 1); DataCo_Source_System_Discovery.pdf (Phase 2); DataCo_Data_Dictionary.pdf (Phase 3); profiling.ipynb (Phase 4); cleaning.ipynb (Phase 5); 00_Mental_Model_of_Data_Modeling.md and 01_business_model.md (Phase 6); 01–07 SQL scripts (Phases 7–8); 07_00_Business_Question_Map.md (Phase 9).

D.5 Glossary

Grain — The level of detail at which a fact table is recorded. In this project, the grain is order-line item.

Dimension — A table providing business context (customer, product, date).

Fact — A table recording business events and measurements (fact_sales).

Star schema — A data model with one central fact table surrounded by dimension tables.

Forensics — The assessment of whether analytical findings can be trusted as evidence about a real business.

Synthetic data — Data constructed for research or testing purposes, not recorded from real business activity.

DATA CO SUPPLY CHAIN ANALYTICS

End-to-End Analytics & Data Forensics Investigation

END OF DOCUMENT

End of Appendix D. End of Final Report.